Semester Project Report

Visualizing US border-crossing behavior since 1996

(1) Dataset

The dataset for the project is the US border crossing data, which contains all the incidents of crossing the border into the US, provided by the Bureau of Transportation Statistics, Govt. of the US. This dataset tells about the incoming border crossing counts into the US.

Below is a preview of the raw data:

	Port Name	State	Port Code	Border	Date	Measure	Value	Location
0	Calexico East	California	2507	US-Mexico Border	03/01/2019 12:00:00 AM	Trucks	34447	POINT (-115.48433000000001 32.67524)
1	Van Buren	Maine	108	US-Canada Border	03/01/2019 12:00:00 AM	Rail Containers Full	428	POINT (-67.94271 47.16207)
2	Otay Mesa	California	2506	US-Mexico Border	03/01/2019 12:00:00 AM	Trucks	81217	POINT (-117.05333 32.57333)
3	Nogales	Arizona	2604	US-Mexico Border	03/01/2019 12:00:00 AM	Trains	62	POINT (-110.93361 31.340279999999996)
4	Trout River	New York	715	US-Canada Border	03/01/2019 12:00:00 AM	Personal Vehicle Passengers	16377	POINT (-73.44253 44.990010000000005)

And below is a detailed description of the dataset schema:

	Variable	Explanation	type	example
1	Port Name	Name of the port	string	Van Buren
2	State	Name of the state	string	Maine
3	Port Code	Code of the port	integer	108
4	Border	Type of border(US-Mexico/US-Canada)	category	US-Mexico Border
5	Date	Date for the border crossing incident	date	03/01/2019 12:00:00 AM
6	Measure	Measure of border crossing	category	Trucks
7	Value	Number of border crossing incidents	integer	81217
8	Location	longitude and latitude of the port	fields	POINT (-67.94271 47.16207)

(2) Data preprocessing

Some data preprocessings are conducted before further analysis and virtualization. Two features are generated for each observation:

- Day of the week (numeric)
- Is week day (bool)

And more data was joined to further explore and visualize some potential reasons that might affect border-crossing behavior, including.

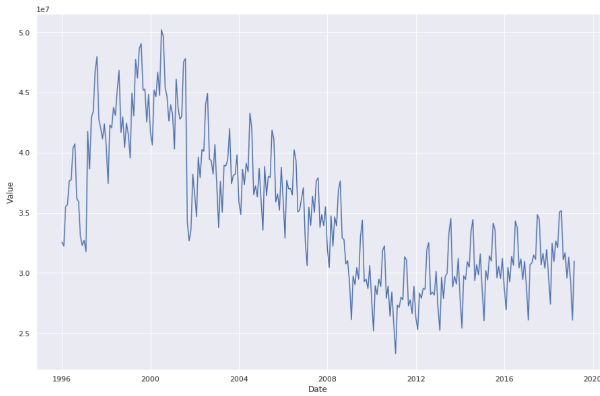
- GDP stats of each state
- Average temperature of each state per month

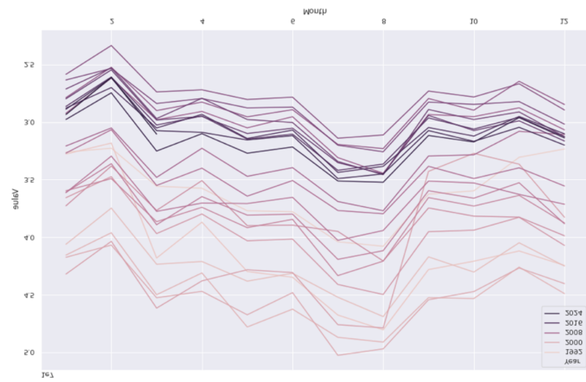
Besides, standardization and one-hot encoding were also applied on the dataset, respectively on numeric and categorical features.

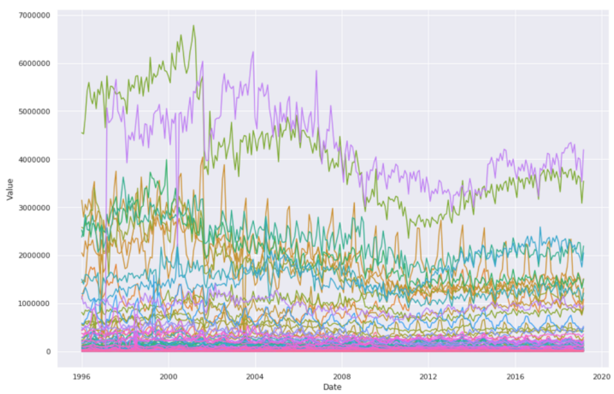
(3) Visualizations

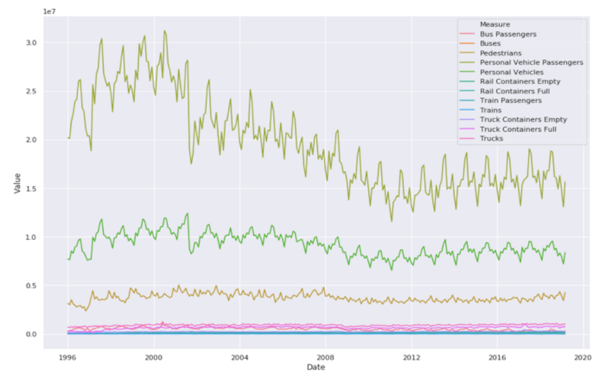
Here're the detailed visualizations on our analysis.

First, below is the charts describing the trend of cross-border count going with time, with breakdown in year, port and measure (approach).





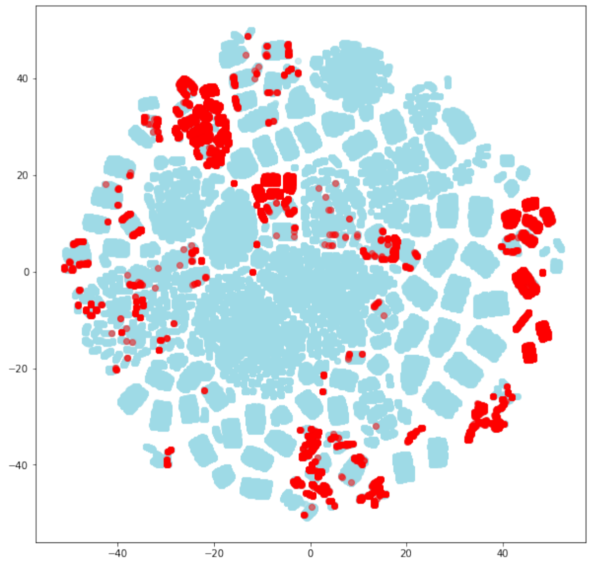




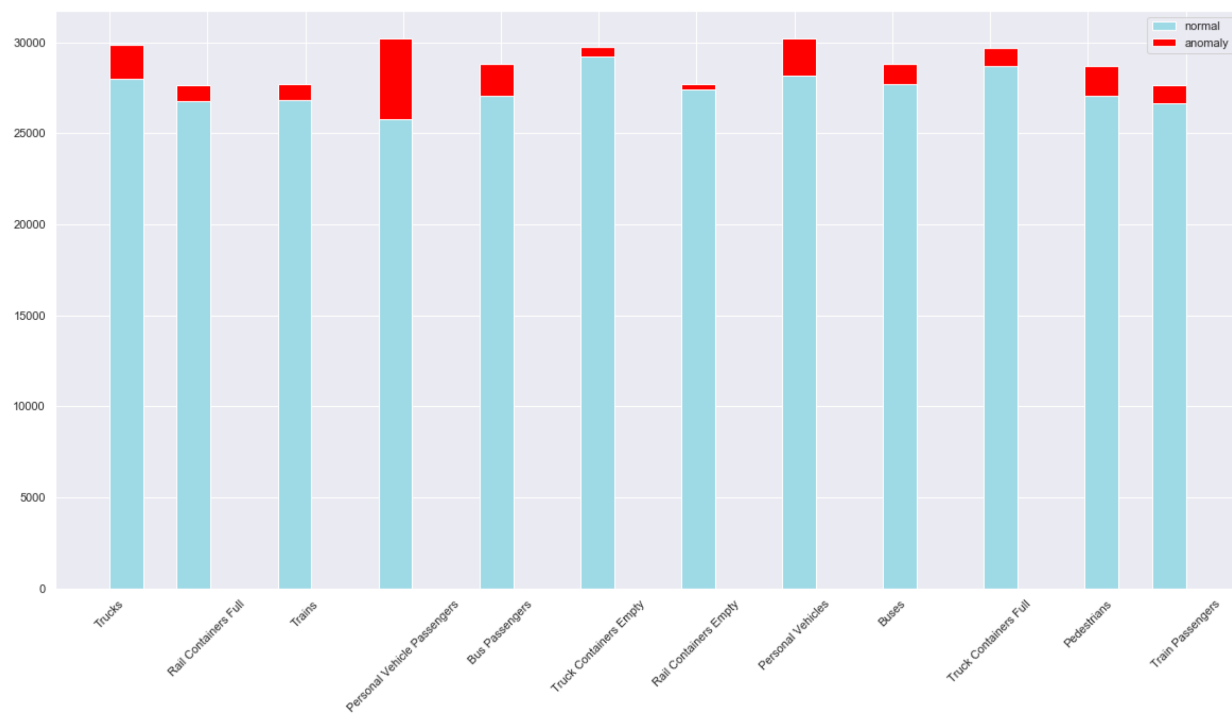
To better compare the crossing counts under different approaches, each measure is plotted and combined into a complete plot.

Then, anomaly detection was used to find the outliers in the data, and visualizations were further applied to illustrate the anomaly detection process.

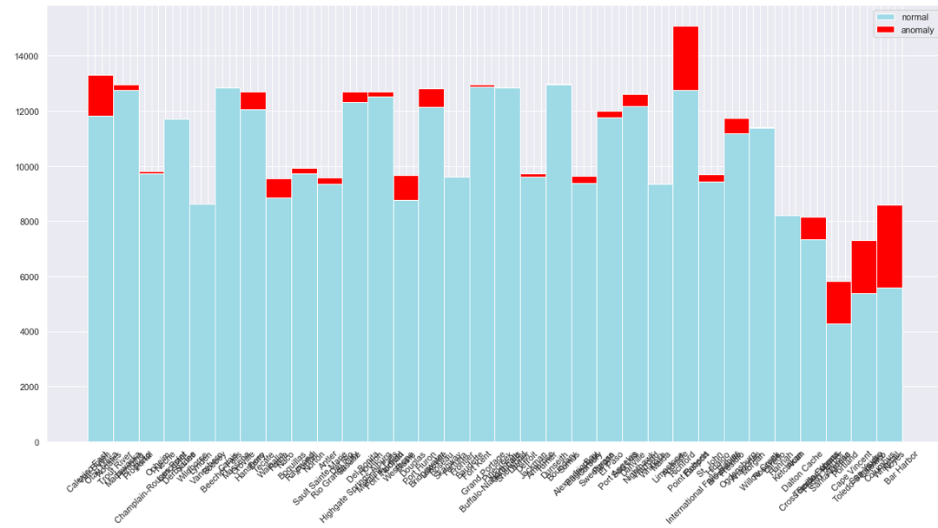
After detecting anomalies using isolation forest, dimensionality reduction techniques were used to enable a 2-d plot on the anomalies. The 2-d scatter plots after applying t-SNE dimensionality reduction is plotted as below, where blue dots are normal ones and red ones are anomalies.



For the distribution of anomalies in our data, the below charts display the number of anomalies detected in each category(measure and port name, respectively).

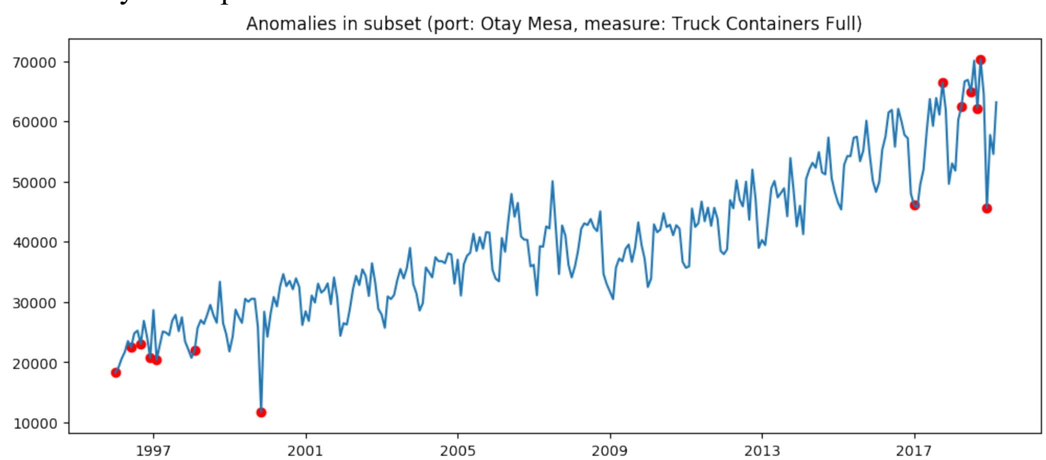


Anomalies detected in each measure

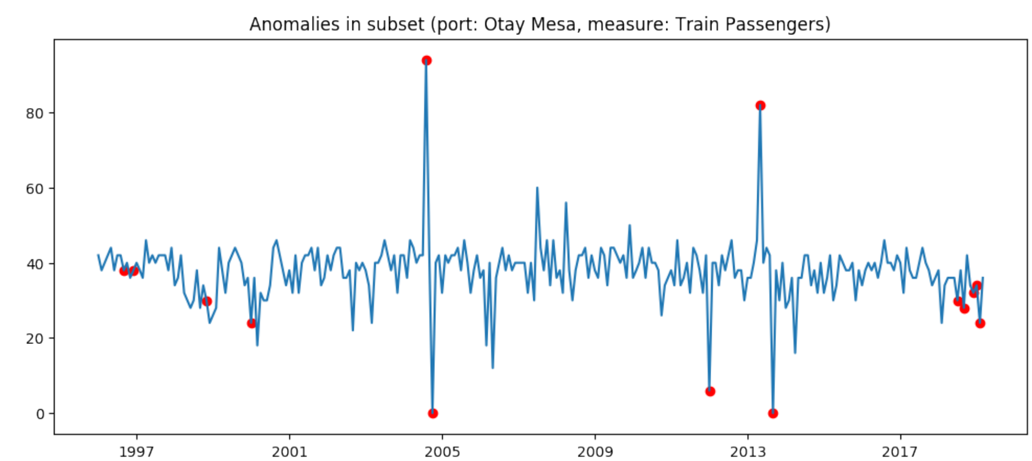


Anomalies detected in each port

And more specifically, the anomaly values detected are also visualized. For example, the next chart plots both normal values (blue) with anomaly values (red) inside the subset of truck measure and Otay Mesa port.



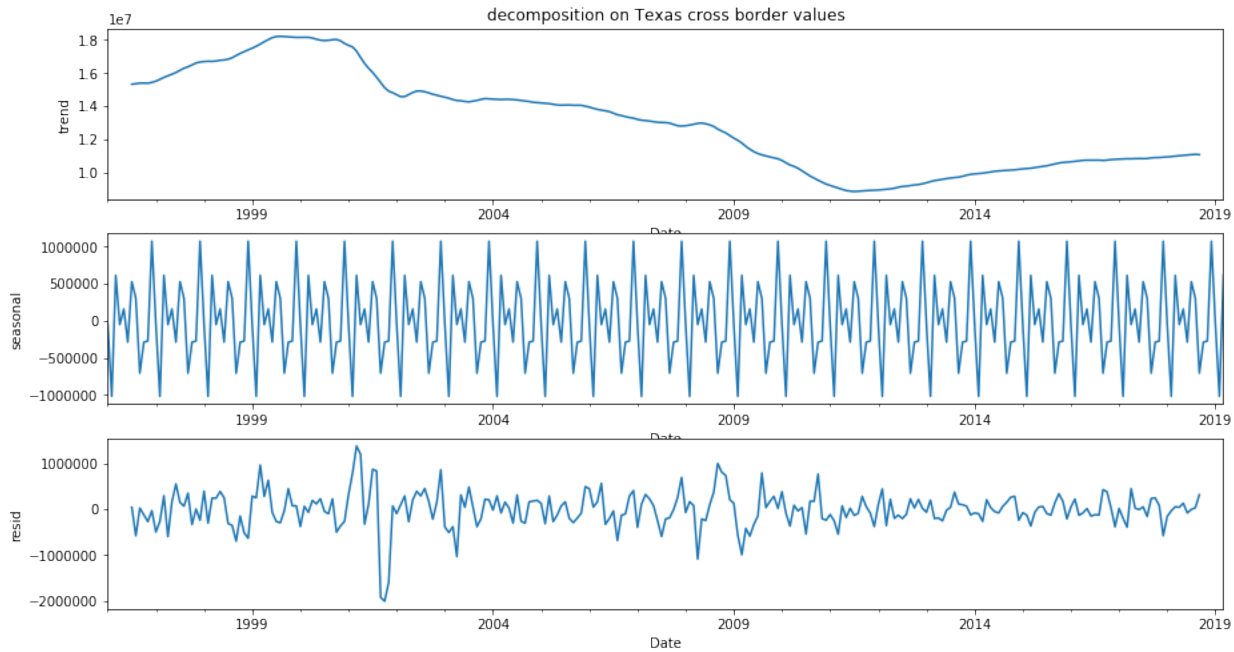
The below chart shows the same visualization on the subset of train measure and Otay Mesa port.



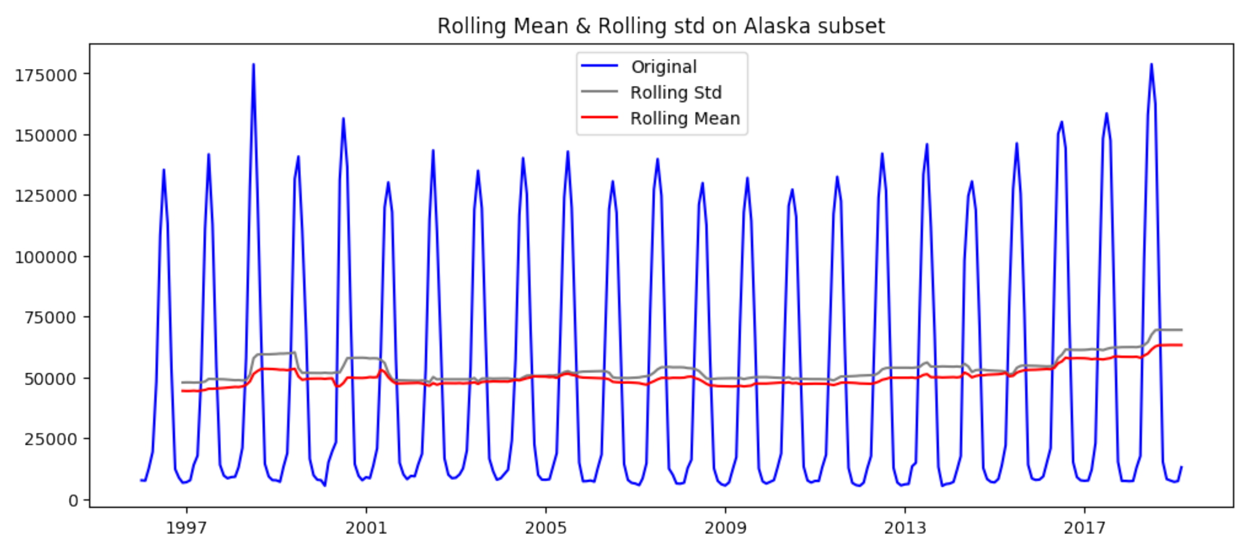
It could be easily observed that most anomalies detected by isolation forest tend to be spike/trough values.

Finally for exploration, a 3-d plot on the anomaly distribution was plotted for the California subset, which can be viewed at: http://people.ku.edu/~c693s270/. This is an interactive plot which supports interaction such as zoom in/out, dragging and rotation to view the full scattered dot details.

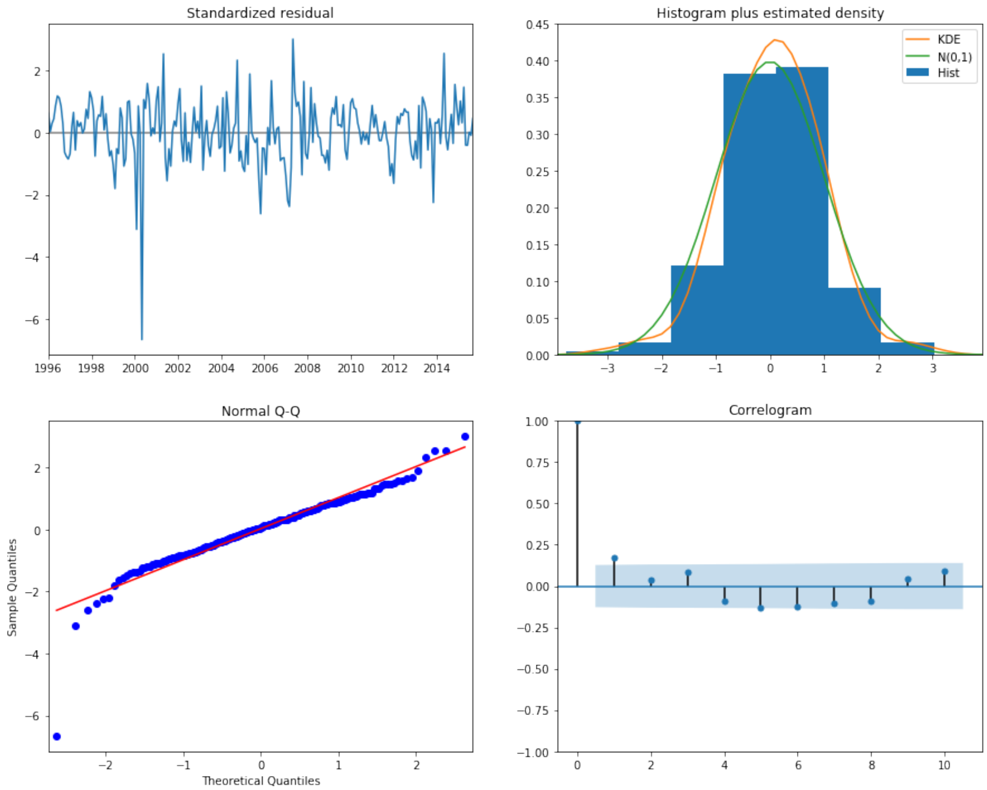
The last part is a time series analysis based on the datetime column we have. Below is a decomposition analysis on the Texas subset values.



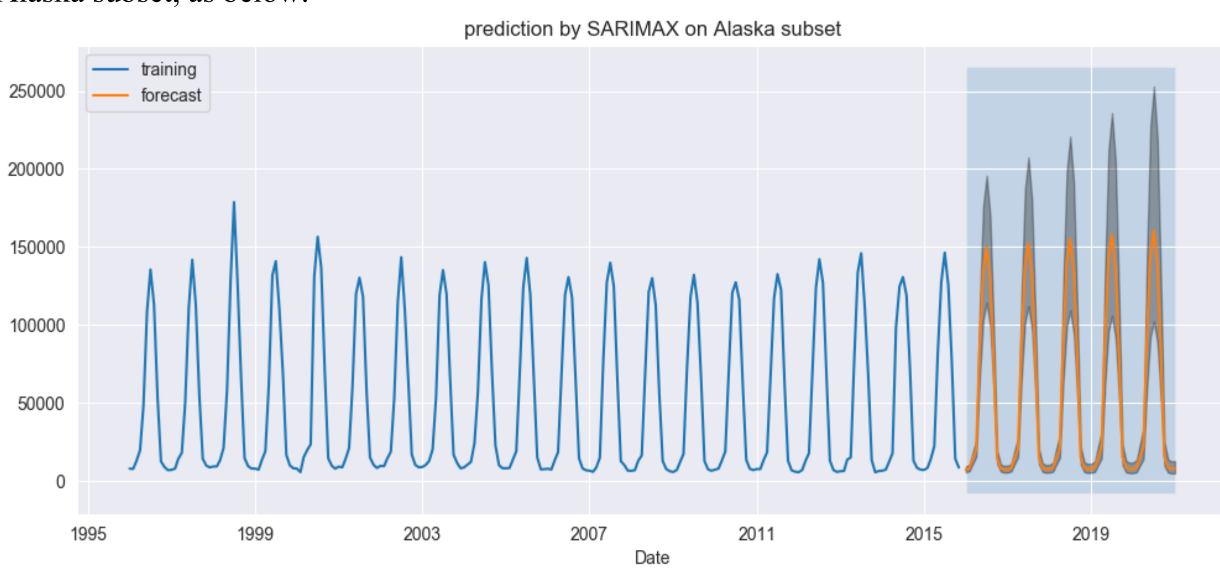
An evident seasonality pattern could be observed after decomposing trend from periodicity and residual. To better verify our analysis, the Alaska subset was further chosen, which could help reason our visualization results. Below is the original value trend and its rolling stats going over time.



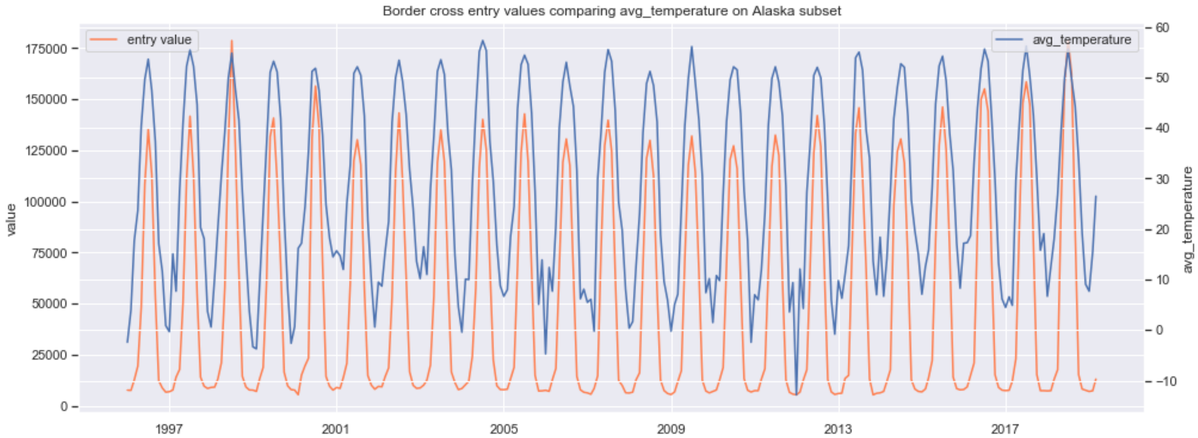
Besides a strong seasonality, its stable rolling mean and std demonstrate little disturbance from factors other than temperature.
More time series analysis was further explored. The below visualization illustrates the effectiveness of fitting the Alaska subset data using a seasonal ARIMA model.



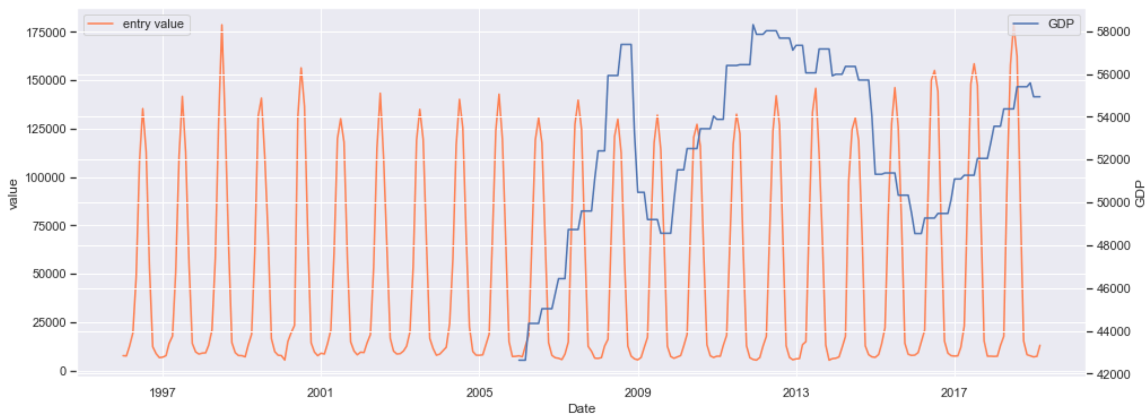
These 4 plots illustrate the distribution of residuals and density, as well as quantile distribution and autocorrelation, which demonstrate a rather good performance of data fitting, thus showing a strong correlation between cross-border values and seasonality.
Going further from that, the seasonal ARIMA model was used to predict future values for the Alaska subset, as below.



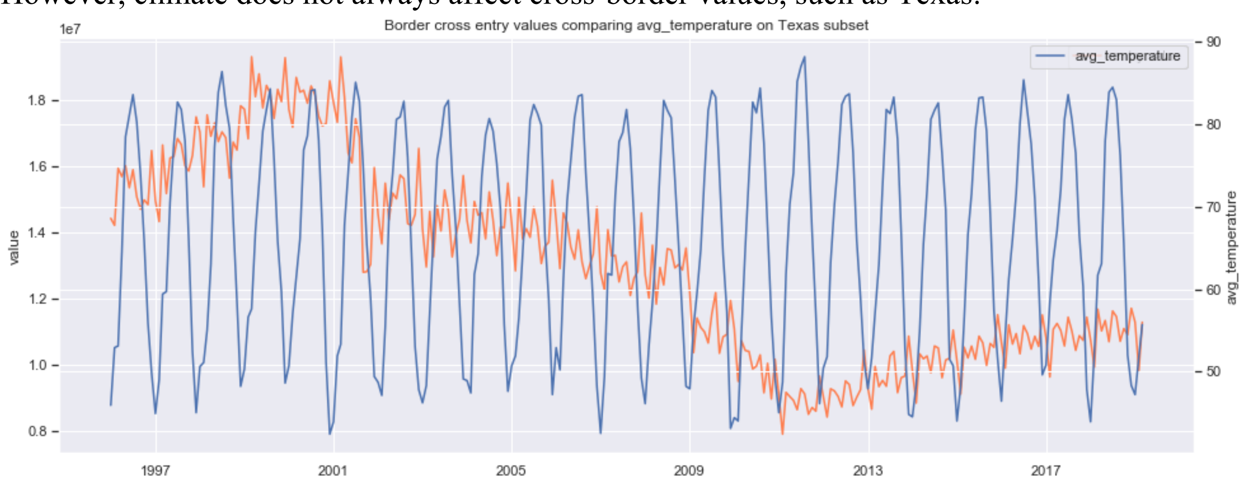
Finally, two comparisons were made to explore if there exists any correlation between cross-border value and other factors, which are the two external datasets we joined.
The next chart compares the average temperature in Alaska and the border crossing values.



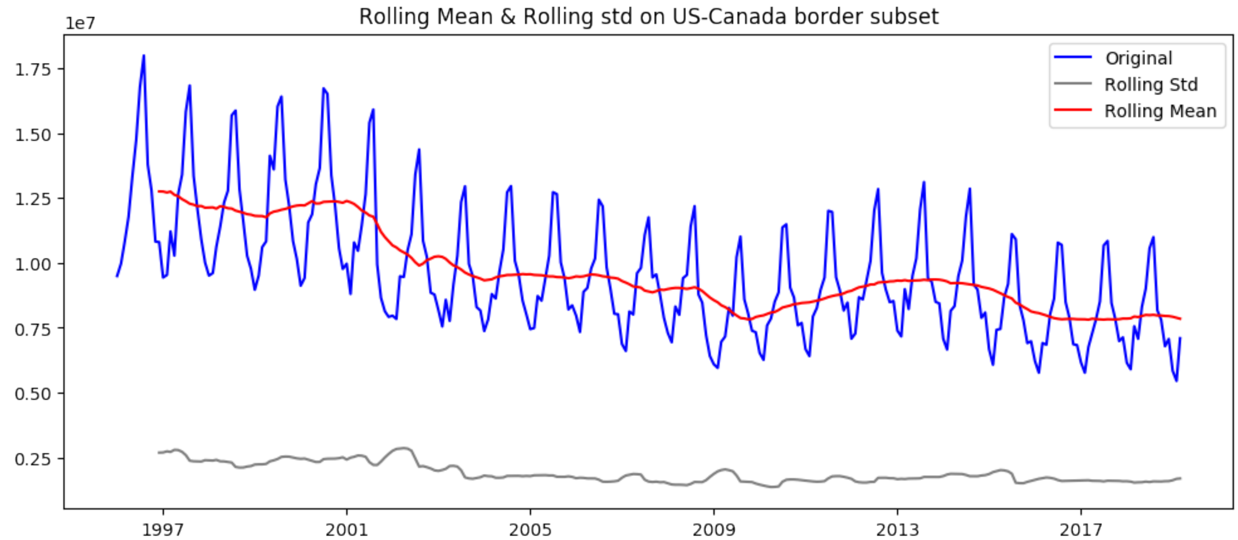
It's obvious that they're strongly correlated, which almost seems to be the only reason affecting cross-border quantities in Alaska.
Then, a similar comparison between cross-border value and state GDP was plotted.



No available data was found before 2005, so the GDP data starts from 2005. This time, no evidence can be observed to support the idea that economic factors would influence cross-border behavior in Alaska.
However, climate does not always affect cross-border values, such as Texas:



And, the US-Canada border subset seems to be affected by not only temperatures. Additional trends are observed besides repetitive seasonality. That's quite reasonable since the US-Canada border consists of quite a few ports that could involve far more excessive factors than Alaska has.



In summary, the dataset of US border crossing behavior is processed, analyzed and visualized in several aspects such as descriptive stats, anomaly detection and time series analysis. We explored and found several interesting facts on border-crossing behavior, and deep dived several typical subsets to visualize their anomalies as well as results of applying time series modeling.